

DEON YEBOAH TAKYI-AMPONSEM

Portfolio: <https://deontakyi.vercel.app/> Github: <https://github.com/Calibre-sys>

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PROFESSIONAL SUMMARY

Self-taught software developer and published machine-learning researcher. I build AI-enabled products end to end, from unsupervised machine learning systems, published in a peer-reviewed journal, to full-stack ML web applications, to workflow automations.

TECHNICAL SKILLS

- **Languages:** Python (Fast API), JavaScript (Node.js, React.js, Google Apps Script), HTML/CSS, VBA
- **Machine Learning & Data Science:** supervised & unsupervised learning, deep learning, time-series forecasting, MLOps
- **Automation & Tooling:** Google Apps Script, Googleapis, Zapier, AI agents Development, financial modelling with VBA/Excel
- **Applied LLMS:** MCPs, Context Windows, RAG.

EXPERIENCE

Business Development / Technical Associate (National Service) | SOH Energy, Ghana Nov 2025 – Present

- Built SOH Order Management system, a software that streamlines BDC operations by automating document generation, and email dispatch, tracking of orders, cutting processing time by over 90%.
- Built an AI-agent recruitment system that screens and ranks CVs automatically, replacing manual shortlisting and generating candidate emails end to end.
- Replaced a 900-contact manual phone-call campaign with a fully automated Google Apps Script email workflow — custom emails with one-click responses, automatic sheet logging, 3-day follow-up reminders, and thank-you confirmations, converting 455 of 880 contacts (52%) with zero manual calls.

Reservoir Characterization Lead (ML Research) | i-DPRS Research Lab, Ghana Sep 2024 – Present

- Built RescharAI, an industry software with Fast API that applies unsupervised ML for HFU identification and supervised ML with residual correctors for net-pay delineation.
- Collaborated with Ghana National Petroleum Corporation (GNPC) to apply unsupervised machine learning to enhance reservoir characterization on real field data.
- Engineered a causal analysis tool for production engineers to determine the effects of water injectors on producers, improving

SELECTED PROJECTS

SOH Order Tracker - Automated document software integrating Google Sheets and automated emails, reducing the processing time for orders and invoices by over 90%.

RescharAI — ML software for reservoir characterization

- Designed and built software combining unsupervised learning (HFU identification) and supervised learning with residual correctors (net-pay delineation), removing the need for manual petrophysical workflows.

Fortel — no-code machine-learning web application (www.fortel.onrender.com)

- Co-developed a full-stack ML web app that lets users train models on securely uploaded private data without writing code

PUBLICATION

- <https://doi.org/10.1016/j.jappgeo.2025.105778>
- <https://doi.org/10.2118/233427-MS>